
RL Fine-Tuning of Language Models

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1 Introduction

This project will explore RL fine-tuning of the language model with an emphasis on instruction following.

2 Related Work

Language model finetuning has been studied in prompting, fine-tuning, test-time inference, and reinforcement learning.

2.1 Prompting

Prompting improves model reasoning without changing the model. Wei et al. (2022b) provides chain of thought demonstrations as exemplars in prompting and LLM shows reasoning capability and accuracy is improved. (Chen et al. (2023)) replaces the chain of thoughts prompting by program of thoughts prompting and shows better performance in reasoning tasks. The weakness of prompting without changing the model is that if the model wasn't pretrained with relevant task or knowledge, the model may not perform well.

2.2 Finetuning

Fine-tuning resumes model training on task-specific dataset to align model to the tasks. Fine-tuning can be considered as imitation learning or behavior cloning. Wei et al. (2022a) shows finetuning model on a dataset with instruction improves zero-shot performance on unseen tasks. Imitation learning suffers from covariate shift and compounding error. DAgger (Dataset Aggregation) can be applied to mitigate the problem. Touvron et al. (2023) and Bai et al. (2022) applies rejection sampling fine-tuning, a DAgger solution that iteratively fine-tunes with new expert ranked data. In each iteration, the model generates K outputs. K outputs are ranked by a reward model and the highest ranked output is fed to the next fine-tuning iteration. There is also research Xiong et al. (2025) suggesting rejection sampling fine-tuning or also called reward ranked fine-tuning can achieve competitive performance with policy gradient method such as PPO. Traditionally, fine-tuning or imitation learning is considered to be unable to surpass the expert.

2.3 Test time inference

Wang et al. (2023) uses test time inference to generate multiple outputs parallelly and pick the best output by self-consistency among outputs.

2.4 Reinforcement learning

Ouyang et al. (2022) applies reinforcement learning to LLM with PPO. ? shows reinforcement learning alone without fine-tuning can produce high quality reasoning capability.

3 Method

3.1 Data

Structured conversation datasets are processed in the following steps:

1. Truncated to just the first round of conversation between user and assistant.
2. Convert structured conversation to a string using the QWen chat template. QWen uses special tokens to separate system prompt, user prompt, and assistant response.

Example chat template

```
<|im_start|>system
You are a helpful assistant.<|im_end|>
<|im_start|>user
Hello, how are you?<|im_end|>
<|im_start|>assistant
I'm doing great. How can I help you today?<|im_end|><|endoftext|>
```

3. The string is tokenized by the QWen tokenizer with maximum length of 768. Conversations longer than 768 tokens are discarded. Conversations shorter than 768 tokens are padded at the right. Positions of assistant response are marked by the assistant mask. The assistant mask will later be used to limit loss computation to only assistant response. The chat template doesn't automatically add end of text token at the end. End of text token is explicitly added after the assistant response. Then end of text token is considered part of the assistant response and included in loss computation.

3.2 SFT

Supervised finetuning (SFT) optimizes for the next token prediction task. Only prediction in assistant response is used for loss computation. SFT clones the behavior of the dataset.

$$objective = \max_{\theta} \mathbb{E}_{x, y \in D} \sum_{t=1}^{|y|} \log \pi_{\theta}(y_t \mid x, y_{<t})$$

3.3 DPO

DPO is an offline RL algorithm. DPO models the reward of a response as the log ratio between sequence probability from policy model and sequence probability from reference model. DPO formulates the loss function as the Bradley-Terry objective, which takes the reward of chosen response and the reward of rejected response and uses the reward difference to classify whether one response is better than the other response.

Loss function:

$$\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right]$$

β controls how much KL penalty is applied. Larger β means less deviation from the reference model.

SFT model is used as baseline model in DPO.

3.4 Extension 1

Extension 1 tried multiple adjustment to DPO loss function.

1. Change sequence probability to mean token probability

2. Penalize negative movement. Not only encourage chosen reward to be larger than rejected reward, but also encourage chosen response reward (the log ratio) to be larger than 0, and rejected response reward to be lower than 0.

3.5 Extension 2

Make DPO online RL by iteratively training DPO and revising training data.

Training data revise process:

1. Sample N user prompts from ultrafeedback dataset.
2. Synthesize response for sampled user prompts by latest model
3. Compare synthesized response with previous chosen response in the sample by nemotron reward model
4. If synthesized response is better than the previous chosen response, change the chosen response to the synthesized response, and change the rejected response to the previous chosen response. If synthesized response is worse than the previous chosen response, update rejected response to the synthesized response.

Training iteration process:

1. for iteration 0, revise training data by baseline model; for iteration 1 and forward, revise training data by last DPO model
2. create a new DPO model by the revised training data. back to 1 and repeat

4 Experimental Setup

Datasets:

1. SmolTalk Allal et al. (2025) for SFT. SmolTalk contains a set of conversations between user and expert assistant.
2. Ultrafeedback Dubois et al. (2024) for DPO and extension. Ultrafeedback contains win and loss response to the same set of user prompts.

Evaluation metrics:

Model inference is ran by VLLM. Model response is sent to nemotron reward model. The reward score is used to determine whether model response is better than reference model.

For SFT, the evaluation metric is the win rate over base qwen model.

For DPO, the evaluation metric is the win rate over SFT model.

For extension, the evaluation metric is the win rate over DPO.

SFT, DPO, and extension are also evaluated by the heldout prompts on leaderboard.

Training setup:

1. 1024 sequence length.
2. 1 L40S gpu on g6e xlarge
3. torch.compile for speed
4. float32 model weight with automatic mixed precision of bfloat16
5. Gradient accumulation every 4 steps and gradient checkpointing to reduce ram
6. full parameter finetuning
7. 1 epoch on full train split
8. optimizer is AdamW 0.1 weight decay

9. SFT: batch size is 16. Initial learning rate is $7e-6$. Learning rate linear decay to 10% of initial
10. DPO: batch size is 8. Initial learning rate is $1e-6$. Learning rate linear warmup from 0 in first 160 batches and linear decay to 20% of initial
11. Gradient norm clip to 1.0

Generation config:

1. temperature = 0.6
2. top k = 20
3. top p = 0.95
4. besides the eos token, <lim_endl> is also used as stop generation token

5 Results

5.1 Quantitative Evaluation

SFT win rate over base model: 73%

SFT final leaderboard score: 0.115

DPO win rate over SFT: 53%

DPO final leaderboard score: 0.1825

Extension 1 was tried but results were lost. My impression was extension 1 was neutral with regard to DPO. Due to time limit, I move forward with extension 2.

Extension 2 results:

Extension win rate over DPO: 51%

Extension final leaderboard score: leaderboard was overwhelmed by too many submissions. submitted extension response json to gradescope instead.

Summary

Metric shows DPO generates better response than SFT.

5.2 Qualitative Analysis

5.2.1 SFT vs. base

User prompt: You will be given a definition of a task first, then some input of the task. In this task, you are given a text of news article and corresponding headline of an article. Your task is to give label "match" if headline is correct for article, otherwise give label "no".

Article: MORE than 100 pints of blood were donated in one hour yesterday by Embu town residents for victims of the Westgate Mall terrorist attack. Headline: 100 pints of blood donated in one hour

SFT model response: match

Base model response: "That is a great task you have assigned," said a Kentucky Board of Education administrator writing

Summary SFT follows instruction well, while base fails to follow instruction.

5.2.2 DPO vs SFT

User prompt:

Detailed Instructions: In this task, you're given the title of a five-sentence story and the first four sentences. Your job is to write the last sentence of the story such that it seamlessly connects with the rest of the story.

Problem: Title: The Sore Back. Sentence 1: Julian had been suffering from back pains all week. Sentence 2: It was beginning to interfere with his work. Sentence 3: As he tried to find the cause, he noticed his bed was very uniform. Sentence 4: He decided to replace his mattress that very day.

Solution:

SFT model response: Julian sat in his chair, feeling the ache in his back grow stronger with each passing day. As he sat there, he couldn't help but think about the countless hours he'd spent in bed, nursing his back.

DPO model response:

As he carefully replaced the old mattress with a new, custom-fit one, Julian's back pain finally began to ease, allowing him to focus on his work without the constant discomfort.

Summary DPO response is more cohesive with the context.

Extension vs DPO

User prompt:

Given the matrix as $A = \begin{bmatrix} 2 & -3 \\ 5 & 6 \end{bmatrix}$, find the determinant

DPO model response: To find the determinant of the given matrix $A = \begin{bmatrix} 2 & -3 \\ 5 & 6 \end{bmatrix}$, we can use the formula for the determinant of a 2x2 matrix. The determinant of a matrix $A = [a_{ij}]$ is given by the expression $a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$.

For matrix $A = \begin{bmatrix} 2 & -3 \\ 5 & 6 \end{bmatrix}$, we have $a_{11} = 2$, $a_{12} = -3$, $a_{21} = 5$, $a_{22} = 6$, and $a_{23} = 0$. Plugging these values into the formula, we get:

$$\text{Determinant}(A) = (2)(6*0 - 0*6) - (-3)(5*0 - 0*5) + (-3)(5*6 - (-3)*5) = (2)(0) - (-3)(0) + (-3)(30 + 15) = 0 + 0 + (-3)(45) = -135.$$

So, the determinant of matrix A is -135.

Extension model response: To find the determinant of a 2x2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the determinant is calculated as $\det(A) = ad - bc$.

Given matrix $A = \begin{bmatrix} 2 & -3 \\ 5 & 6 \end{bmatrix}$, we can substitute the values into the formula.

$$\det(A) = (2)(6) - (-3)(5) = 12 + 15 = 27.$$

The determinant of matrix A is 27.

Summary Extension response is accurate while DPO response is wrong.

6 Team Contributions

- **Yukang Yang:** data processing, model training of sft, dpo, extensions, model inference, evaluation, poster, presentation, report.

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